

Mode 10 — Square-BFP (centered / opt)

C-DP8 (C8×C8) → DP16 (8×8) → DP16 (8×4) → DP8 (8×4) → block exponents → DP8 (4×4) → centered add-and-square

Step 1 — C-DP8 (C8×C8)

$$\text{C-DP8 (C8} \times \text{C8)} = D = \sum_{i=0}^7 a_i b_i, \quad a_i = a_i^{re} + j a_i^{im}, \quad b_i = b_i^{re} + j b_i^{im}$$

Step 2 — DP16 (8×8)

$$\text{Re}(D) = \underbrace{\sum_{i=0}^7 a_i^{re} b_i^{re} - \sum_{i=0}^7 a_i^{im} b_i^{im}}_{\text{DP16 (8} \times \text{8)}}$$

$$\text{Im}(D) = \underbrace{\sum_{i=0}^7 a_i^{re} b_i^{im} + \sum_{i=0}^7 a_i^{im} b_i^{re}}_{\text{DP16 (8} \times \text{8)}}$$

Step 3 — DP16 (8×4)

$$\text{Re}(D) = 2^4 \underbrace{\sum_{i=0}^7 a_i^{re} b_i^{re,hi} - \sum_{i=0}^7 a_i^{im} b_i^{im,hi}}_{\text{DP16 (8} \times \text{4)}} + \underbrace{\sum_{i=0}^7 a_i^{re} b_i^{re,lo} - \sum_{i=0}^7 a_i^{im} b_i^{im,lo}}_{\text{DP16 (8} \times \text{4)}}$$

$$\text{Im}(D) = 2^4 \underbrace{\sum_{i=0}^7 a_i^{re} b_i^{im,hi} + \sum_{i=0}^7 a_i^{im} b_i^{re,hi}}_{\text{DP16 (8} \times \text{4)}} + \underbrace{\sum_{i=0}^7 a_i^{re} b_i^{im,lo} + \sum_{i=0}^7 a_i^{im} b_i^{re,lo}}_{\text{DP16 (8} \times \text{4)}}$$

Step 4 — DP8 (8×4)

$$\text{Re}(D) = 2^4 \underbrace{\sum_{i=0}^7 a_i^{re} b_i^{re,hi}}_{\text{DP8 (8} \times \text{4)}} - 2^4 \underbrace{\sum_{i=0}^7 a_i^{im} b_i^{im,hi}}_{\text{DP8 (8} \times \text{4)}} + \underbrace{\sum_{i=0}^7 a_i^{re} b_i^{re,lo}}_{\text{DP8 (8} \times \text{4)}} - \underbrace{\sum_{i=0}^7 a_i^{im} b_i^{im,lo}}_{\text{DP8 (8} \times \text{4)}}$$

$$\text{Im}(D) = 2^4 \underbrace{\sum_{i=0}^7 a_i^{re} b_i^{im,hi}}_{\text{DP8 (8} \times \text{4)}} + 2^4 \underbrace{\sum_{i=0}^7 a_i^{im} b_i^{re,hi}}_{\text{DP8 (8} \times \text{4)}} + \underbrace{\sum_{i=0}^7 a_i^{re} b_i^{im,lo}}_{\text{DP8 (8} \times \text{4)}} + \underbrace{\sum_{i=0}^7 a_i^{im} b_i^{re,lo}}_{\text{DP8 (8} \times \text{4)}}$$

Step 5 — Block exponents (one exponent block per DP8 (8×4) — all copies of one E ; re/im tied, negation scale-blind; parallel matmul: own E_a, E_b)

$$(P_j, E) \text{ per DP8 (8} \times \text{4) above,} \quad E = E_a + E_b \longrightarrow (M^{re}, E), \quad (M^{im}, E)$$

$\delta \equiv 0$: M^{re}, M^{im} are the Step-4 sums unchanged; $\text{Re}(\hat{D}) = M^{re} \cdot 2^E$, $\text{Im}(\hat{D}) = M^{im} \cdot 2^E$.

Step 6 — DP8 (4×4)

$$\begin{aligned} M^{re} = & 2^8 \underbrace{\sum_{i=0}^7 a_{H,i}^{re} b_i^{re,hi}}_{\text{DP8 (4} \times \text{4)}} + 2^4 \underbrace{\sum_{i=0}^7 a_{L,i}^{re} b_i^{re,hi}}_{\text{DP8 (4} \times \text{4)}} - 2^8 \underbrace{\sum_{i=0}^7 a_{H,i}^{im} b_i^{im,hi}}_{\text{DP8 (4} \times \text{4)}} - 2^4 \underbrace{\sum_{i=0}^7 a_{L,i}^{im} b_i^{im,hi}}_{\text{DP8 (4} \times \text{4)}} \\ & + 2^4 \underbrace{\sum_{i=0}^7 a_{H,i}^{re} b_i^{re,lo}}_{\text{DP8 (4} \times \text{4)}} + \underbrace{\sum_{i=0}^7 a_{L,i}^{re} b_i^{re,lo}}_{\text{DP8 (4} \times \text{4)}} - 2^4 \underbrace{\sum_{i=0}^7 a_{H,i}^{im} b_i^{im,lo}}_{\text{DP8 (4} \times \text{4)}} - \underbrace{\sum_{i=0}^7 a_{L,i}^{im} b_i^{im,lo}}_{\text{DP8 (4} \times \text{4)}} \end{aligned}$$

$$\begin{aligned}
M^{im} = & \underbrace{2^8 \sum_{i=0}^7 a_{H,i}^{re} b_i^{im,hi}}_{\text{DP8 (4x4)}} + \underbrace{2^4 \sum_{i=0}^7 a_{L,i}^{re} b_i^{im,hi}}_{\text{DP8 (4x4)}} + \underbrace{2^8 \sum_{i=0}^7 a_{H,i}^{im} b_i^{re,hi}}_{\text{DP8 (4x4)}} + \underbrace{2^4 \sum_{i=0}^7 a_{L,i}^{im} b_i^{re,hi}}_{\text{DP8 (4x4)}} \\
& + \underbrace{2^4 \sum_{i=0}^7 a_{H,i}^{re} b_i^{im,lo}}_{\text{DP8 (4x4)}} + \underbrace{\sum_{i=0}^7 a_{L,i}^{re} b_i^{im,lo}}_{\text{DP8 (4x4)}} + \underbrace{2^4 \sum_{i=0}^7 a_{H,i}^{im} b_i^{re,lo}}_{\text{DP8 (4x4)}} + \underbrace{\sum_{i=0}^7 a_{L,i}^{im} b_i^{re,lo}}_{\text{DP8 (4x4)}}
\end{aligned}$$

Step 7 — Centered add-and-square (subtract 8 from unsigned nibbles; per DP8 (4x4) add $c = 0/512/2048$ for 0/1/2 unsigned nibbles. The $-\Sigma a^{im}b^{im}$ group keeps the add-square but carries a global $-$.)

$$\begin{aligned}
M^{re} = & \underbrace{2^8 \frac{1}{2} \left(\sum_{i=0}^7 (a_{H,i}^{re} + b_i^{re,hi})^2 - \sum_{i=0}^7 (a_{H,i}^{re})^2 - \sum_{i=0}^7 (b_i^{re,hi})^2 \right)}_{\text{DP8 (4x4)}} \\
& + \underbrace{2^4 \frac{1}{2} \left(\sum_{i=0}^7 ((a_{L,i}^{re} - 8) + b_i^{re,hi})^2 - \sum_{i=0}^7 (a_{L,i}^{re} - 8)^2 - \sum_{i=0}^7 (b_i^{re,hi} - 8)^2 + 512 \right)}_{\text{DP8 (4x4)}} \\
& - \underbrace{2^8 \frac{1}{2} \left(\sum_{i=0}^7 (a_{H,i}^{im} + b_i^{im,hi})^2 - \sum_{i=0}^7 (a_{H,i}^{im})^2 - \sum_{i=0}^7 (b_i^{im,hi})^2 \right)}_{\text{DP8 (4x4)}} \\
& - \underbrace{2^4 \frac{1}{2} \left(\sum_{i=0}^7 ((a_{L,i}^{im} - 8) + b_i^{im,hi})^2 - \sum_{i=0}^7 (a_{L,i}^{im} - 8)^2 - \sum_{i=0}^7 (b_i^{im,hi} - 8)^2 + 512 \right)}_{\text{DP8 (4x4)}} \\
& + \underbrace{2^4 \frac{1}{2} \left(\sum_{i=0}^7 (a_{H,i}^{re} + (b_i^{re,lo} - 8))^2 - \sum_{i=0}^7 (a_{H,i}^{re} - 8)^2 - \sum_{i=0}^7 (b_i^{re,lo} - 8)^2 + 512 \right)}_{\text{DP8 (4x4)}} \\
& + \underbrace{\frac{1}{2} \left(\sum_{i=0}^7 ((a_{L,i}^{re} - 8) + (b_i^{re,lo} - 8))^2 - \sum_{i=0}^7 (a_{L,i}^{re} - 16)^2 - \sum_{i=0}^7 (b_i^{re,lo} - 16)^2 + 2048 \right)}_{\text{DP8 (4x4)}} \\
& - \underbrace{2^4 \frac{1}{2} \left(\sum_{i=0}^7 (a_{H,i}^{im} + (b_i^{im,lo} - 8))^2 - \sum_{i=0}^7 (a_{H,i}^{im} - 8)^2 - \sum_{i=0}^7 (b_i^{im,lo} - 8)^2 + 512 \right)}_{\text{DP8 (4x4)}} \\
& - \underbrace{\frac{1}{2} \left(\sum_{i=0}^7 ((a_{L,i}^{im} - 8) + (b_i^{im,lo} - 8))^2 - \sum_{i=0}^7 (a_{L,i}^{im} - 16)^2 - \sum_{i=0}^7 (b_i^{im,lo} - 16)^2 + 2048 \right)}_{\text{DP8 (4x4)}} \\
\\
M^{im} = & \underbrace{2^8 \frac{1}{2} \left(\sum_{i=0}^7 (a_{H,i}^{re} + b_i^{im,hi})^2 - \sum_{i=0}^7 (a_{H,i}^{re})^2 - \sum_{i=0}^7 (b_i^{im,hi})^2 \right)}_{\text{DP8 (4x4)}} \\
& + \underbrace{2^4 \frac{1}{2} \left(\sum_{i=0}^7 ((a_{L,i}^{re} - 8) + b_i^{im,hi})^2 - \sum_{i=0}^7 (a_{L,i}^{re} - 8)^2 - \sum_{i=0}^7 (b_i^{im,hi} - 8)^2 + 512 \right)}_{\text{DP8 (4x4)}} \\
& + \underbrace{2^8 \frac{1}{2} \left(\sum_{i=0}^7 (a_{H,i}^{im} + b_i^{re,hi})^2 - \sum_{i=0}^7 (a_{H,i}^{im})^2 - \sum_{i=0}^7 (b_i^{re,hi})^2 \right)}_{\text{DP8 (4x4)}}
\end{aligned}$$

$$\begin{aligned}
& + 2^4 \frac{1}{2} \underbrace{\left(\sum_{i=0}^7 ((a_{L,i}^{im} - 8) + b_i^{re,hi})^2 - \sum_{i=0}^7 (a_{L,i}^{im} - 8)^2 - \sum_{i=0}^7 (b_i^{re,hi} - 8)^2 + 512 \right)}_{\text{DP8 (4x4)}} \\
& + 2^4 \frac{1}{2} \underbrace{\left(\sum_{i=0}^7 (a_{H,i}^{re} + (b_i^{im,lo} - 8))^2 - \sum_{i=0}^7 (a_{H,i}^{re} - 8)^2 - \sum_{i=0}^7 (b_i^{im,lo} - 8)^2 + 512 \right)}_{\text{DP8 (4x4)}} \\
& + \frac{1}{2} \underbrace{\left(\sum_{i=0}^7 ((a_{L,i}^{re} - 8) + (b_i^{im,lo} - 8))^2 - \sum_{i=0}^7 (a_{L,i}^{re} - 16)^2 - \sum_{i=0}^7 (b_i^{im,lo} - 16)^2 + 2048 \right)}_{\text{DP8 (4x4)}} \\
& + 2^4 \frac{1}{2} \underbrace{\left(\sum_{i=0}^7 (a_{H,i}^{im} + (b_i^{re,lo} - 8))^2 - \sum_{i=0}^7 (a_{H,i}^{im} - 8)^2 - \sum_{i=0}^7 (b_i^{re,lo} - 8)^2 + 512 \right)}_{\text{DP8 (4x4)}} \\
& + \frac{1}{2} \underbrace{\left(\sum_{i=0}^7 ((a_{L,i}^{im} - 8) + (b_i^{re,lo} - 8))^2 - \sum_{i=0}^7 (a_{L,i}^{im} - 16)^2 - \sum_{i=0}^7 (b_i^{re,lo} - 16)^2 + 2048 \right)}_{\text{DP8 (4x4)}}
\end{aligned}$$

Step 8 — All components

Re (M^{re}): (the $a^{im}b^{im}$ blocks carry a global $-$ in PE, α , β and C)

$$\begin{aligned}
\text{PE} &= 2^8 \sum_{i=0}^7 (a_{H,i}^{re} + b_i^{re,hi})^2 + 2^4 \sum_{i=0}^7 ((a_{L,i}^{re} - 8) + b_i^{re,hi})^2 - 2^8 \sum_{i=0}^7 (a_{H,i}^{im} + b_i^{im,hi})^2 - 2^4 \sum_{i=0}^7 ((a_{L,i}^{im} - 8) + b_i^{im,hi})^2 \\
&+ 2^4 \sum_{i=0}^7 (a_{H,i}^{re} + (b_i^{re,lo} - 8))^2 + \sum_{i=0}^7 ((a_{L,i}^{re} - 8) + (b_i^{re,lo} - 8))^2 - 2^4 \sum_{i=0}^7 (a_{H,i}^{im} + (b_i^{im,lo} - 8))^2 - \sum_{i=0}^7 ((a_{L,i}^{im} - 8) + (b_i^{im,lo} - 8))^2 \\
\alpha &= 2^8 \sum_{i=0}^7 (a_{H,i}^{re})^2 + 2^4 \sum_{i=0}^7 (a_{L,i}^{re} - 8)^2 - 2^8 \sum_{i=0}^7 (a_{H,i}^{im})^2 - 2^4 \sum_{i=0}^7 (a_{L,i}^{im} - 8)^2 \\
&+ 2^4 \sum_{i=0}^7 (a_{H,i}^{re} - 8)^2 + \sum_{i=0}^7 (a_{L,i}^{re} - 16)^2 - 2^4 \sum_{i=0}^7 (a_{H,i}^{im} - 8)^2 - \sum_{i=0}^7 (a_{L,i}^{im} - 16)^2 \\
\beta &= 2^8 \sum_{i=0}^7 (b_i^{re,hi})^2 + 2^4 \sum_{i=0}^7 (b_i^{re,hi} - 8)^2 - 2^8 \sum_{i=0}^7 (b_i^{im,hi})^2 - 2^4 \sum_{i=0}^7 (b_i^{im,hi} - 8)^2 \\
&+ 2^4 \sum_{i=0}^7 (b_i^{re,lo} - 8)^2 + \sum_{i=0}^7 (b_i^{re,lo} - 16)^2 - 2^4 \sum_{i=0}^7 (b_i^{im,lo} - 8)^2 - \sum_{i=0}^7 (b_i^{im,lo} - 16)^2 \\
C &= 0 + 8192 + 0 - 8192 + 8192 + 2048 - 8192 - 2048 = 0 \\
M^{re} &= \frac{1}{2} (\text{PE} - \alpha - \beta + C), \quad E = E_a + E_b
\end{aligned}$$

Im (M^{im}):

$$\begin{aligned}
\text{PE} &= 2^8 \sum_{i=0}^7 (a_{H,i}^{re} + b_i^{im,hi})^2 + 2^4 \sum_{i=0}^7 ((a_{L,i}^{re} - 8) + b_i^{im,hi})^2 + 2^8 \sum_{i=0}^7 (a_{H,i}^{im} + b_i^{re,hi})^2 + 2^4 \sum_{i=0}^7 ((a_{L,i}^{im} - 8) + b_i^{re,hi})^2 \\
&\quad + 2^4 \sum_{i=0}^7 (a_{H,i}^{re} + (b_i^{im,lo} - 8))^2 + \sum_{i=0}^7 ((a_{L,i}^{re} - 8) + (b_i^{im,lo} - 8))^2 + 2^4 \sum_{i=0}^7 (a_{H,i}^{im} + (b_i^{re,lo} - 8))^2 + \sum_{i=0}^7 ((a_{L,i}^{im} - 8) + (b_i^{re,lo} - 8))^2 \\
\alpha &= 2^8 \sum_{i=0}^7 (a_{H,i}^{re})^2 + 2^4 \sum_{i=0}^7 (a_{L,i}^{re} - 8)^2 + 2^8 \sum_{i=0}^7 (a_{H,i}^{im})^2 + 2^4 \sum_{i=0}^7 (a_{L,i}^{im} - 8)^2 \\
&\quad + 2^4 \sum_{i=0}^7 (a_{H,i}^{re} - 8)^2 + \sum_{i=0}^7 (a_{L,i}^{re} - 16)^2 + 2^4 \sum_{i=0}^7 (a_{H,i}^{im} - 8)^2 + \sum_{i=0}^7 (a_{L,i}^{im} - 16)^2 \\
\beta &= 2^8 \sum_{i=0}^7 (b_i^{im,hi})^2 + 2^4 \sum_{i=0}^7 (b_i^{im,hi} - 8)^2 + 2^8 \sum_{i=0}^7 (b_i^{re,hi})^2 + 2^4 \sum_{i=0}^7 (b_i^{re,hi} - 8)^2 \\
&\quad + 2^4 \sum_{i=0}^7 (b_i^{im,lo} - 8)^2 + \sum_{i=0}^7 (b_i^{im,lo} - 16)^2 + 2^4 \sum_{i=0}^7 (b_i^{re,lo} - 8)^2 + \sum_{i=0}^7 (b_i^{re,lo} - 16)^2 \\
C &= 0 + 8192 + 0 + 8192 + 8192 + 2048 + 8192 + 2048 = 36864 \\
M^{im} &= \frac{1}{2} (\text{PE} - \alpha - \beta + C), \quad E = E_a + E_b
\end{aligned}$$